

NiCE Search Grid Change History Guide

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==          Search Grid 20.x Change History          ==
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== IMPORTANT NOTE:                                     ==
== Pay particular attention to items marked as "Deprecated" or "Removed". ==
== Deprecated items are likely to be fully removed in a future release. ==
== Always read README.txt before installing or upgrading Search Grid. ==
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Search Grid 20.x is the same branch as Search Grid 3.2, and 12.4, just renumbered as part of a grid-wide numbering scheme change. (What would have been 3.2.2 became 12.4.2, and then later, what would have been 12.4.102 became 20.0.0.) Therefore, this 20.x change history includes all changes back through version 3.2.0.

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Version 20.1.2
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Administration and Deployment

[NXQC-43991] Added get-gridResourceUtilization management-console command, which will report CPU and memory usage for every gateway, compute node, and data node in the grid. A browser can also be used to query this info by going to:
<http://<GatewayIP>:25002/about/gridResourceUtilization>
 and logging in with the admin password (same as management-console's).

Security hardening & Technical Debt

* [NXQCC-64874] Updated the following components:
 - Logback to 1.5.25 to eliminate two security vulnerabilities per

Black Duck:

BDSA-2025-12973 (medium risk)

BDSA-2026-0927/CVE-2026-1225 (low risk)

- Azul Zulu to the latest available, 17.64.16 (jre 17.0.18).

* [NXQC-64633] This ticket includes the following changes:

- Updated components to eliminate security vulnerabilities reported by Black Duck. (Some components were updated only because they were transitive dependencies of components that were needed.)

- jakarta-annotations 2.1.1. (Added.)

- zookeeper to 3.9.4, eliminating one high-risk security vulnerability: BDSA-2025-12456.

- netty to 4.2.9.Final.

- netty-tcnative to 2.0.70.Final, eliminating 5 medium-risk security vulnerabilities:

BDSA-2025-10732 (CVE-2025-58057)

BDSA-2025-13945

BDSA-2025-10730 (CVE-2025-58056)

BDSA-2025-8614 (CVE-2025-55163)

BDSA-2025-62789 (CVE-2025-67735).

- artemis to 2.44.0 (no vulnerabilities).

- jakarta-jms-api to 3.1.0 (no vulnerabilities).

- apache curator to 5.9.0 (no vulnerabilities).

- gridproducts-platform to 1.1.1 (no vulnerabilities).

- Updated gridproductrs-cep-client to 2.0.1 which includes an enhancement to work around an unlikely corner-case condition caused by Artemis that could have caused Search Grid to not start, or not shut down, or appear hung, IF AND ONLY IF Search Grid was configured to use CEP, and while Artemis on the CEP server was in a bad state where it refused messages, but in a way where the Artemis client library would loop internally retrying rather than returning an error. The most likely cause of this behavior as if the CEP server's used disk space was above the Artemis threshold (default 90%). NOTE: If that Artemis server issue were to happen now, with this work-around, Search Grid could still occasionally appear sluggish (and a lot will be logged), but nothing will hang.

- Eliminated an erroneous and misleading log message from the Control Service that was logged any time data nodes were restarted or their connectivity was lost and restored, incorrectly reporting that data nodes' names were not in the nodeNameList.

Bug Fixes

None

General

None

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Version 20.1.1

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Administration and Deployment

None

Security hardening

- * [NXQC-59801] Updated workbench to 10.13.4.100, which fixes an issue where a particular type of media file corruption could cause a hang. This version also potentially improves memory usage and performance of phrase recognition for some non-English phrase packs when a GPU is being used.
- * [NXQC-61225] Addressed vulnerabilities identified by a penetration test, related to Derby and mbeans. Also updated the following components:
 - Updated the Azul JRE to 17.60.18 (Java 17.0.16).
 - Updated Artemis to 2.42.0, eliminating medium-risk vulnerability BDSA-2025-4595.
 - Updated commons-lang to 3.18.0 eliminating medium-risk vulnerability BDSA-2025-6881.
 - Updated Jetty to 9.4.58.v20250814 eliminating low-risk vulnerability BDSA-2024-7229.

Bug Fixes

- * [NXQC-60062] Fixed groovy-shell.cmd and groovy-console.cmd to work with Java 17.

General

- * [NXQC-61281] Rebranded NICE to NiCE in all documentation and updated to new logo graphics where applicable.
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Version 20.1.0

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Administration and Deployment

None

Security hardening

* [NXQC-59118] Updated the following components:

- commons-beanutils to 1.11.0
 - * Eliminated one HIGH-risk vulnerability: BDSA-2025-4668
- commons-logging to 1.3.5 (no vulnerabilities, just updating)
- gridproducts-cep-client to 1.4.0 (no vulnerabilities, just updating)

* [NXQC-54196] Updated required Java version to Java 17 as well as updating the following 3rd party libraries, superseding some of the updates mentioned below in NXQC-54757:

- artemis 2.41.0
 - * Eliminated medium-risk security vulnerability BDSA-2025-3090
 - * Eliminated low-risk security vulnerability BDSA-2025-2607
- asn-one 0.6.0 (new component, needed indirectly by artemis)
- commons-logging 1.3.4
- commons-beanutils 1.10.1
- gridproducts-cep-client 1.3.0
- jackson 2.18.4
- jakarta-jms-api 2.0.3
- logback 1.5.18
- pem-keystore 3.0.0
- slf4j 2.0.17

This ticket also disabled the application.wadl HTTP endpoints for security.

* [NXQC-58628] Updated Jetty to 9.4.57.v20241219 to eliminate medium-risk security vulnerability BDSA-2025-3989.

* [NXQC-57863] Updated Netty to 4.2.0.Final to eliminate the following security vulnerabilities:

- * Medium: BDSA-2025-1033
- * Medium: BDSA-2025-1031 (CVE-2025-25193)

* [NXQC-57854] Updated Azul Zulu to 11.80.22 jre 11.0.27.

* [NXQC-54757] Updated logback per Black Duck, which necessitated upgrading other 3rd-party components as well, and updated Azul Zulu JRE:

- Updated Logback to 1.5.16, eliminating the following security vulnerabilities:

- * Medium: BDSA-2024-9861
- * Medium: BDSA-2024-9866
- Updated Slf4j to 2.0.16
- Updated Commons-Lang3 to 3.17.0
- Updated Artemis client to 2.38.0
- Updated Jgroups to 5.3.13.Final
- Added pem-keystore 2.4.0 because the new Artemis client needs it.
- Added jdk18on 1.78 because the new Artemis client needs it.
- Updated gridproducts-cep-client to 1.1.1 for compatibility with the updated Logback. (This supersedes v1.0.2 that was part of NXQC-54309 in the Bug Fixes section, below.)
- Updated Azul Zulu JRE to 11.78.16.
- * [NXQC-51840] Updated Jersey to 1.19.4.
- * [NXQC-53586] Updated the following 3rd party components:
 - jackson to v2.18.2.
 - woodstox to 7.0.0.
 - stax2 to 4.2.2.
 - curator to 5.7.1.
 - zookeeper to 3.9.3, eliminating high security vulnerability BDSA-2024-8266, though Search Grid was not susceptible to that vulnerability.
 - netty-tcnative to 2.0.66.
 - commons-io to 2.17.0.
- * [NXQC-53488] Updated the following 3rd party components per Black Duck:
 - Netty upgraded to 4.1.115.Final to eliminate medium security vulnerability BDSA-2024-8565.
 - XStream upgraded to 1.4.21 to eliminate medium security vulnerability BDSA-2024-8322.

----- Bug Fixes

- * [NXQC-54309] Updated gridproducts-cep-client to v1.0.2, which fixes an issue that, ONLY if CEP is enabled, would cause the test-system command to report it didn't get responses from any nodes other than control, ONLY if one or more nodes were actually down. (If all nodes were up, it was fine.)

----- General

- * [NXQC-55874] Refactored code to pull common platform code into a separate library so it can be used by the other grid products.
- * [NXQC-54201] Improve Logging of Initialization and Connectivity Issues.
This includes a new WARN-level gateway log message to gateway if a data

node attempts to register that is not in the nodeNameList, amongst other improvements. Also, the CEP heartbeat messages from the Base service now indicate their leader status (isLeader = true or false).

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Version 20.0.0
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Administration and Deployment

- * [NXQC-43992] Added grid-wide performance logging to the gateway and updated the Search Grid Administrator's guide to document the new log file. Those metrics will also be included in the CEP data that the gateway service sends in its CEP heartbeat if CEP is enabled. (If a grid has multiple gateways, only one will include those metrics in its CEP heartbeat messages.)
- * [NXQC-46994] Added protection of grid data in the event that the control service's data is wiped or corrupted. This also includes other related changes:
 - Base and/or Gateway services will intentionally fail if invalid control node data is detected at start-up, in order to preserve the integrity of data on all data nodes. (If this were to happen, manually initiated control node data reconstruction will need to be performed, per the Search Grid Administration Guide.)
 - Improvements were made to the reconstruction of control data, including clarifications in the "Control Node Reconstruction" section of the Search Grid Administration Guide.
 - Resource leaks in the "restart-system" management-console command were eliminated. (This command is used internally when reconstructing control node data.)
 - The "test-system" management-console command's "-verbose" option has been enhanced to make post-reconstruction validation a little easier.
 - The Admin Guide has been updated to include improved, detailed instructions for adding or removing control nodes, and a script has been added to aid in this.
- * [NXQC-46283] Implemented Common Exception Protocol (CEP) support. (Disabled by default. Requires CEP Server product in order to make use of this new feature.) Updated Search Grid Administration Guide accordingly.
- * [NXQC-47861] Enhanced the management-console to provide more meaningful information if run from the command-line with improper parameters, and exits with a non-zero error code if there are ANY errors now, including command line usage errors and command file errors.
- * [NXQC-49634] Documented Disaster Recovery critical limitation that requires

the backup grid instance to have only 1 active gateway. (Please refer to the Disaster Recovery section of the NiCE Search Grid Administration Guide.) This effectively disables NXPCG-1988 added in 12.4.100. The primary instance can still specify multiple IP addresses in the backupInstanceSystemUri property, however, only the first one listed will be used for now.

Also, improved some DR error logging and the DR metrics.

- * [NXQC-45737] Added CPU utilization to the test-system details (-verbose) output and updated Admin Guide to mention the Service User's permission required for CPU utilization be reported (otherwise reports "Unavailable").
- * [NXQC-43918] Added a new, "Moving Search Grid's Folders" section to the Search Grid System Administration Guide, which details the process required in order to not break Search Grid nor break the next NxDeployer upgrade to Search Grid.

Security hardening

- * [NXQC-53278] Upgraded Groovy to 3.0.22.
- * [NXQC-52815] Upgraded Azul Zulu JRE to 11.76.22.
- * [NXQC-52702] Updated Apache Avro to v1.12.0 per Black Duck to eliminate one HIGH vulnerability: BDSA-2024-6954. (NOTE: Search Grid was immune to that vulnerability because it uses embedded schemas in both the server and client libraries, but we're upgrading as a matter of best-practice.)
- * [NXQC-52306] Updated the following 3rd party components:
 - Netty to 4.1.113.Final, which eliminates one HIGH vulnerability, BDSA-2024-0720 (CVE-2024-29025).
 - Jackson to 2.18.0.
 - Curator to 5.7.0.
 - Error-prone-annotations to 2.21.1.
 - Checker-qual to 3.37.0.
- * [NXQC-51767] Updated the following 3rd party components:
 - Jetty to 9.4.56.v20240826
 - Mimepull to 1.10.0
- * [NXQC-51277] Updated Spring to 5.3.39 to eliminate 3 high vulnerabilities:
 - BDSA-2024-1160 (CVE-2024-22262)
 - BDSA-2024-0402 (CVE-2024-22243)
 - BDSA-2024-0625 (CVE-2024-22259)and 2 medium vulnerabilities:
 - BDSA-2024-5369
 - BDSA-2024-5371
- * [NXQC-45046] Updated Zookeeper per Black Duck:
 - Zookeeper has been updated to 3.9.2 to eliminate a medium security vulnerability: BDSA-2024-0640/CVE-2024-23944.

- * [NXQC-44383] Updated components per Black Duck to eliminate security vulnerabilities:
 - Apache commons-compress has been updated to 1.26.0 eliminate two potential medium security vulnerabilities, BDSA-2024-0360/CVE-2024-26308 and BDSA-2024-0363/CVE-2024-25710. (Search Grid was not directly vulnerable, but updated to latest code anyway as best-practice.)
 - Apache commons-io has been updated to 2.15.1 as a recursive dependency of Apache commons-compress.
 - Apache commons-lang3 has been updated to 3.14.0 as a recursive dependency of Apache commons-compress.
 - Apache commons-text 1.11.0 has been added as a replacement for something that used be in the older commons-lang3
 - Apache commons-rng 1.5 has been added as a replacement for something that used to be in the older commons-lang3

Bug Fixes

- * [NXQC-52383] Fixed an issue that would prevent the control service from being stopped if stopping it was attempted before the service finished initializing and was fully up.
- * [NXQC-52303] Fixed an issue that could cause the control service to not start if the grid had multiple control nodes and one was started without any of the others started within a minute or so. (This bug had existed since Search Grid v3.2.0.)
- * [NXQC-50123] Fixed corner case condition that could prevent data nodes from registering with gateway. (Highly unlikely in a production environment.)
- * [NXQC-47043] Ported bug fix NXQC-47015 from 12.4.101 patch branch into 20.0.0. Note: That bug has never been in ANY released version of 20.0.x. It is mentioned here so that it's clear that 20.0.x doesn't contain that issue.

General

- * [NXQC-52249] Retry failed deletion attempts of temporary files when the cause is that the file "is being used by another process", rather than just logging the failure. (This benign, intermittent issue has existed for at least ten years. This change avoids the issue by retrying deletion attempts, which then succeeds.) NOTE: Old temp*.pat files in the <data location>\temp folder that have accumulated from older versions of Search Grid may be removed MANUALLY. Any temp*.pat file more than two minutes old is safe to be deleted.
- * [NXQC-44643] Upgraded Workbench to 10.12.1.7 and set SG's default Workbench

logging to "Warn" level to take advantage of that new Workbench feature.

- * [NXQC-47116] Cleaned up some logging to improve readability.
- * [NXQC-43213] Enhanced public Get Media Inventory API to optionally allow querying for only media that were declared during a specified time-span.
- * [NXQC-45669] Updated all HTML documents to improve formatting when they get imported into DocHub. Also fixed a few typos that were noticed.
- * [NXQC-45030] Updated Workbench to 10.11.0.2 and implemented a means to optionally output workbench logging for certain testing scenarios. NOTE: This new Workbench version also has an optimization that could improve phrase recognition performance when GPUs are being used, but it is a very specific issue that was addressed relating to the GPU waiting on a highly over-burdened CPU to give it more work, and therefore that optimization is unlikely to affect most customer installations.
- * [NXQC-43243] Added more specific logging to indicate which DR properties are misconfigured if replication fails on the backup instance.
- * [NXQC-44403] Corrected a miscalculation in determining the number of threads that can be used on the CPU to handle overflow of work from the GPU if overflowing is enabled (it is disabled by default).

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Version 12.4.100
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Administration and Deployment

- * [NXPCG-2116] Removed Windows Server 2012 R2 from the list of supported platforms in the Search Grid Requirements document, and explained this in README.txt.
- * [NXPCG-2001] Disaster Recovery no longer requires keeping the default admin password. Changing the password on the primary requires manually updating it on the backup.
- * [NXPCG-1988] Added support for multiple gateways in Disaster Recovery's backup instance by allowing comma-separated values in the nexidia.searchgrid.disasterrecovery.backupInstanceSystemUri property on the primary instance.
- * [NXPCG-1292] Modified data redistribution to no longer require maintenance mode, thus allowing it to be run while Search Grid is live.

Security hardening

- * [NXPCG-2111] Updated Azul Zulu JRE to v11.70.16 (Java v11.0.22)

- * [NXPCG-2077] Updated Logback to v1.2.13 to eliminate 2 medium security vulnerabilities: BDSA-2023-3341 (CVE-2023-6481) and BDSA-2023-3307 (CVE-2023-6378).
- * [NXPCG-2012] Updated Netty to 4.1.103.Final per Black Duck, eliminating one high vulnerability (BDSA-2023-2732/CVE-2023-44487) and one medium vulnerability (BDSA-2018-4022).
- * [NXPCG-2016] Updated Azul Zulu JRE to 11.68.18 (Java 11.0.21).
- * [NXPCG-1941] Eliminated several vulnerabilities identified by Black Duck:
 - (a) Updated Avro to 1.11.3 to eliminate a medium security vulnerability (BDSA-2023-2603 / CVE-2023-39410)
 - (b) Updated NewtonSoft.Json.dll for searchgrid-client to 13.0.3, which eliminates a security weakness, CWE-755, though there is no CVE associated with this issue.
 - (c) Updated Netty to 4.1.100.Final to eliminate one high security vulnerability (BDSA-2023-2732 / CVE-2023-44487), and one medium security vulnerability (BDSA-2018-4022)
 - (d) Updated snappy-java to 1.1.10.5 to eliminate a medium security vulnerability (BDSA-2023-2536 / CVE-2023-43642).
 - (e) Updated jackson-databind to 2.15.3 to eliminate a potential medium security vulnerability (BDSA-2023-1491 / CVE-2023-35116), though Black Duck says that vulnerability is unconfirmed to be an actual issue.
 - (f) Updated Apache Zookeeper to 3.9.1 to eliminate a medium security vulnerability (BDSA-2023-2742 / CVE-2023-44981).
 - (g) Updated Jetty to 9.4.53.v20231009 to eliminate one high security vulnerability (BDSA-2023-2732 / CVE-2023-44487) and one medium security vulnerability (BDSA-2023-2721).
- * [NXPCG-1852] Search Grid has been migrated from Java 8 to Java 11 (Azul 11.66.20 jre 11.0.20.1), also includes changes to many third-party libraries:
 - activation 1.1.1 -> 1.2.0
 - asm REMOVED
 - commons-codec 1.15 -> 1.16.0
 - commons-compress 1.21 -> 1.24.0
 - groovy 2.4.14 -> 3.0.19
 - httpclient 4.5.13 -> 4.5.14
 - httpcore 4.5.15 -> 4.5.16
 - httpmime 4.5.13 -> 4.5.14
 - jaxb 2.2.6 -> 2.3.1
 - jetty 9.4.51.v20230217 -> 9.4.52.v20230823
 - logback 1.2.10 -> 1.2.12
 - mimepull 1.8 -> 1.9.3
 - netty 4.1.96.Final -> 4.1.97.Final
 - spring 5.3.27 -> 5.3.29

- woodstox-core 6.4.0 -> 6.5.1
- fastinfoset 1.2.15 (newly required)
- istack-commons-runtime 3.0.7 (newly required)
- stax-ex 1.8 (newly required)

Bug Fixes

- * [NXQC-51958] Fixed critical issue with Disaster Recovery. All installations using Disaster Recovery prior to upgrading to Search Grid 20.x, PLEASE SEE README.txt FOR ADDITIONAL, CRITICAL INFORMATION.
- * [NXPCG-2110] Updated to Workbench 10.9.5.34 to pull two Workbench bug fixes:
 - NXR-5211: The issue was rare, but manifested itself by rapidly allocating all RAM until it caused a crash during phrase recognition, but IF AND ONLY IF a media file had a DC-offset during periods of silence.
 - NXR-5664: There was a potential for a crash in ntdll.dll due to workbench not checking a corner-case condition before a memcpy. This was seen at only one site, and only when using a 10.9 Phrase Pack, but it was crashing very frequently when performing phrase recognition with that Phrase Pack.

General

- * [NXPCG-1984] Updated the HTML documents to show the Nexidia logo on browser tabs.
- * [NXPCG-1912] Minor logging improvement of an exception condition.

Version 12.4.4

Administration and Deployment

- * [NXPCG-1807] Improved the Administration Guide by retitling the "Data Redistribution" section to "Adding or Decommissioning Data Nodes & Data Redistribution" and clarifying instructions for decommissioning a node within that section, plus added new "Sever Migration" section.

Security Hardening

- * [NXPCG-1899] Upgraded Guava to 32.1.2 per Black Duck, in order to resolve

one "medium" security vulnerability: BDSA-2016-1748 (CVE-2023-2976). This also automatically updated several dependencies of Guava to newer versions: Checker Qual to 3.33.0, error-prone annotations to 2.18.0, and J2ObjC annotations to 2.8.

- * [NXPCG-1849] Upgraded jackson from 2.14.1 to 2.15.2 per Black Duck.
- * [NXPCG-1851] (a) Updated zookeeper to 3.8.2 to better facilitate (b) & (c):
(b) updated snappy-java to 1.1.10.3 to eliminate three "high" security vulnerabilities (CVE-2023-34453, CVE-2023-34454, CVE-2023-34455)
(c) updated netty to 4.1.96.Final (2.0.61.Final for tcnative subcomponents) in order to eliminate one "medium" security vulnerability (BDSA-2023-1556/ CVE-2023-34462).
- (d) Updated Curator (which uses zookeeper) to 5.5.0, just to stick with the latest version, even though there were no vulnerabilities in it.
- * [NXPCG-1858] Updated Azul Zulu JRE 8.72.0.18 (Java 8.0.382)

Bug Fixes

None

General

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- * [NXPCG-1886] Improved logging of Workbench-generated exceptions.
 - * [NXPCG-1871] Improved logging related to searching and Workbench API calls.
 - * [NXPCG-1809] Ensure clear error message logged at service start-up if the service must terminate because it cannot access <data location>\temp.

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Version 12.4.3

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Administration and Deployment

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- * Please see "IMPORTANT UPGRADE PROCEDURE FOR v12.4.3 AND LATER" in README.txt, made necessary by NXPCG-1214, below, in the Security Hardening section.
 - * [NXPCG-1157] Added ability to reconstruct Control Node's data from all the data nodes in the event of a catastrophic failure of Control's server and updated the Administration Guide accordingly.
 - * [NXPCG-1693] Corrected and clarified the Disaster Recovery section of the

Administration Guide regarding connectivity requirements between the Primary and Backup instances, and clarified the need for identically configured Primary and Backup instance servers.

- * [NXPCG-1770] Clarified and slightly expanded the Upgrades section of the Administration Guide to cover installer GUI behavior for all upgrades.
- * [NXPCG-1657] Updated management-console help output to include a description of common syntax, cleaned-up and slightly enhanced the 'get-help' output, and added "-hidden" and "-brief" options to the "get-help" command. Also changed "test-system" command's "details" option to "-verbose" and changed "get-systemtopolgy" command's "brief" option to "-brief". (Both commands' original option names still work, though are no longer documented.) The Search Grid Administration Guide has been updated accordingly.

----- Security Hardening -----

- * [NXPCG-1715] Updated Azul Zulu JRE to 8.70.0.24 (Java 8.0.372).
- * [NXPCG-1713] Updated Spring to 5.3.27 to eliminate one high vulnerability (BDSA-2023-0649/CVE-2023-20860) and two medium vulnerabilities (BDSA-2023-0638/CVE-2023-20861 and BDSA-2023-0847/CVE-2023-20863), and Updated Jetty to 9.4.51.v20230217 to eliminate two medium vulnerabilities (BDSA-2023-0887 and BDSA-2023-0888).
- * [NXPCG-1630] Upgraded to latest Avro (1.11.1) and its dependencies, eliminating, per Black Duck, the following 4 medium security vulnerabilities: BDSA-2022-2206, BDSA-2022-2207, BDSA-2022-2208, and BDSA-2022-0025 (CVE-2021-43045).
- * [NXPCG-1214] Upgraded to latest Zookeeper, Curator and their dependencies, eliminating, per Black Duck, the following 2 medium security vulnerabilities that were in the older version of Guava required by older Curator: BDSA-2018-1358 (CVE-2018-10237) and BDSA-2018-1358 (CVE-2018-10237).

----- Bug Fixes -----

None

----- General -----

- * [NXPCG-1159] + [NXPCG-1755] Added periodic logging of various performance metrics, which go to separate per-service log files in order to retain these logs beyond when the standard log files roll.
- * [NXPCG-1704] Made a couple of internal/hidden settings dynamically

configurable via global properties to aid in the diagnosis of certain support issues without being intrusive and requiring service restarts.

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Version 12.4.2
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Administration and Deployment

* [NXPCG-1581] Renumbered Search Grid 3.2 to Search Grid 12.4 and updated all pertinent documents. Also corrected a typo in Programmer's Guide that said the max media ID is $2^{62}-1$, when it's really $2^{63}-1$ (signed, positive 64-bit value), and updated client samples "NEWS.txt" to reflect all of the changes between 3.1 and 3.2/12.4, which should have been updated at 3.2.0 but was overlooked.

* [NXPCG-700] Updated Workbench to 10.9.1, from which SafeNet licensing has been completely removed, but is otherwise the same as 10.9.0 mentioned below.

* [NXPCG-1431] Updated to Workbench 10.9.0 (superseded by 10.9.1, above) for GPU support for phrase recognition when using version 10.8 phrase packs.

This includes:

- (a) multi-GPU support
- (b) an improvement to the default calculation of phrase recognition
maxConcurrentOperations both with and without GPUs
- (c) an updated "Configuring Phrase Recognition" section of the Admin Guide
- (d) NEW GPU REQUIREMENTS listed in the Search Grid System Requirements document

Additionally the installation space in the Search Grid System Requirements document was updated (which should have happened at 3.2.1, but was missed).

Also included are improvements to memory usage logging and to the "details" option output of the "test-system details" management-console command.

* [NXPCG-1502] Clarify wording for get-structuredquery in Admin Guide.

* [NXPCG-1453] Added service start time to "test-system details" report and improved the detailed output for the Disaster Recovery Backup Gateway to be consistent with the detailed output reported for other node types.

* [NXPCG-1436] The commands "get-sq ALL" and "get-sq COUNT" report all stored structured queries and the total count respectively, while "get-sq <id>" reports a single structured query as it always has.

* [NXPCG-1449] Enhanced Disaster Recovery to accommodate a fail-over test procedure, which has been added to the Administration Guide.

* [NXPCG-1435] The get-pp command now reports the versionedName instead of the Name of the phrase pack.

- * [NXPCG-1382] Search grid no longer uses the service user's temp directory and therefore that location no longer needs to be excluded from antivirus scans.
- * [NXPCG-1211] Enhanced start-replication command to allow a time range, and Improved then maintenance of Disaster Recovery's queue to eliminate the possibility of the metrics (get-drmet) "age" and "latency" being skewed by the timing of queue updates.
- * [NXPCG-1213] Enhanced start-replication command to have an optional data node list parameter and to verify all requested nodes are up before replicating.
- * [NXPCG-1284] Enhanced test-system command to validate the Backup's Search Grid version matches the Primary's and check for unacceptable time difference between the Primary and the Backup.

----- Security Hardening -----

- * [NXPCG-1617] Updated Azul JRE to 8.68.0.22 (Java 8.0.362).
- * [NXPCG-1532] Per Black Duck, updated XStream to 1.4.20, eliminating the following two security vulnerabilities: BDSA-2022-3693/CVE-2022-41966 and BDSA-2022-2580/CVE-2022-40151.
- * [NXPCG-1527] Per Veracode, eliminated a potential XML vulnerability (CWE-611) in Search Grid's MediaHelper class.
- * [NXPCG-1530] Per Black Duck: Updated Woodstox to 6.4.0 and Stax2 to 4.2.1, eliminating the following six medium vulnerabilities: BDSA-2022-2580, BDSA-2022-2582, BDSA-2022-2583, BDSA-2022-2584, BDSA-2022-2585, and BDSA-2022-2586. Also updated Jackson components to 2.14.1 to resolve two medium-level security vulnerabilities (BDSA-2022-2765/CVE-2022-42003 and BDSA-2022-2768/CVE-2022-42004/CWE-400) identified by Black Duck.
- * [NXPCG-1451] Updated Jackson components to 2.13.4 to resolve a medium-level security vulnerability (BDSA-2022-2768/CWE-400) identified by Black Duck.

----- Bug Fixes -----

- * [NXPCG-1553] Fixed all .cmd scripts so they'll now exit with the exit code (errorlevel) that Java returns, not always 0. Always exiting with a zero was accidentally introduced at v3.1.26 and would've caused any customer- or support-supplied scripts that ran Search Grid's scripts (such as running management-console.cmd in scripting mode to perform operations) to think the result was always success, even if there was an error.
- * [NXPCG-1413] Fixed intermittent issue where the control-service could fail to start on the first attempt, when other services are starting in parallel.
- * [NXPCG-1405] Fixed Disaster Recovery backup instance so it can generate

monogests for media that was replicated to it.

General

- * [NXPCG-1593] Potential SQA performance improvement when compute nodes are being used, but the amount of improvement can vary widely (from none to a lot) based upon specific search content and other compute node loading.
- * [NXPCG-1589] Eliminated permanent RAM consumption on Gateway when installing Phrase Packs. This became significant do to the transformer model sizes.

Version 3.2.1

Administration and Deployment

- * [NXPCG-1408] Updated Admin Guide to indicate how to un-set / remove a global property.
- * [NXPCG-1219] The management-console now prevents using certain commands
Disaster Recovery is enabled and not paused.
- * [NXPCG-1365] Added warning in HTTPS support section regarding option checkboxes in Windows' Certificate Import UI.
- * [NXPCG-1369] Updated Azul to 8.64.0.16 (Java 8.0.342).
- * [NXPCG-1337] Updated Workbench to 10.8.0.5 and improved default settings and calculations for maximum concurrent phrase recognition operations. Workbench 10.8.0.5 also fixes a memory leak that was in Workbench 10.6 (in Search Grid versions 3.1.26 through 3.1.30 and 3.2.0) seen at some sites where multiple phrase packs were being used. This update also may slightly change sentiment scores compared to Search Grid 3.2.0 and SG 3.1.30 and earlier. Please see the README.txt file for additional, important details.
- * [NXPCG-1319] Search Grid now sets service recovery properties at installation auto-restart. See README.txt for more detail about these changes.
- * [NXPCG-620] Enhanced test-system command to validate the property "nexidia.searchgrid.control.connectionString" across all nodes, which is especially critical for sites with multiple control nodes.
- * [NXPCG-1285] The test-system command now checks for unacceptable time differences between nodes and reports failure if unacceptable.

Security Hardening (mostly per Black Duck and Veracode)

- * [NXPCG-1344] Per Black Duck, updated Spring to 5.3.21 to resolve two medium security vulnerabilities, BDSA-2022-1330/CVE-2022-22971 and BDSA-2022-1329/CVE-2022-22970, and updated Jetty to 9.4.48.v20220622 to resolve one medium security vulnerability, BDSA-2022-1887, and one low security vulnerability, BDSA-2022-1891.

Bug Fixes

- * [NXPCG-1385] Fixed Disaster Recovery backup instance so that adding and deleting phrase packs can be done when DR is paused, no longer requiring disabling DR and restarting the gateway as it incorrectly required before. Also, the Administration Guide was updated to include the delete-phrasepack command, which was previously omitted in error.

General

- * [NXPCG-1282] Added checking compatibility at start-up between Disaster Recovery primary and backup instances. See README.txt for more information.
- * [NXPCG-1212] For the same externalMediaId on the Disaster Recovery backup instance, a higher id(s) from an incoming bundle will take precedence and all lower IDs found will be marked obsolete.
- * [NXPCG-681] Improved logging (debug & trace levels) for queries and PR.
- * [NXPCG-1295] Reorganized this CHANGELOG.txt to separate out changes that were also in SG 3.1.21 through 3.1.29 to a separate section near the end of this file.

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Version 3.2.0

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Administration and Deployment

- * [NXPCG-1267] Removed Windows Server 2012 as a supported platform from System Requirements doc, but version 2012 R2 remains supported.
- * [NXPCG-1241] Updated Azul to 8.62.0.20 (Java 8.0.332).
- * [NXPCG-1080] Search Grid licenses are no longer required nor supported.
- * [NXPCG-990] Updated Workbench to 10.6.4.26.

Security Hardening (mostly per Black Duck and Veracode)

-
- * [NXPCG-1244] Updated Spring to 5.3.19 per Black Duck.
 - * [NXPCG-1184] Updated the following components per Black Duck scan:
 - jackson-databind to 2.13.2.2 (CVE-2020-36518)
 - jackson (all other components) to 2.13.2
 - * [NXPCG-1123] Updated metrics-core to 3.2.6
 - * [NXPCG-1115] Updated XStream to 1.4.19 per Black Duck (CVE-2021-43859)
 - * [NXPCG-1074] Veracode CWE-611 (XXE vulnerabilities) in new DR code.
 - * [NXPCG-559] Added ACLs to Zookeeper so access is no longer world/anyone.
 - * [NXPCG-64] Eliminated SG Guava usage. (Still needed by 3rd parties.)

REST API and Client

- * [NXPCG-1217] Removed request system mode API previously used to switch between normal and read-only system modes AND removed the 'errorsOccurred' field from the get system mode API reply.
- * [NXPCG-1118] New DR metrics API, which returns a formatted string, not individual items, allowing for easy future info additions without changing the API.
- * [NXPCG-1080] Removed all licensing info & feature info from APIs.
- * [NXPCG-760] Updated Programmer's Guide to highlight API changes.
- * [US18035] Changed to use wide media id's.
- * [US18038] Removed bitset based media id's (media bit vectors).
- * [US19334] Removed term collections.
- * [US14940] Removed ReplicaNodes member of SystemTopologyInfo.
- * [US15782] Removed ability to extract segments from uploaded PAT's and to embed segments in downloaded PAT's.
- * [US13441] Removed MD5 content header in favor of SHA-512 MIC's.
- * [US15510] Removed client key (a.k.a HMAC) support

General

- * [NXPCG-1245] Corrected a few minor typos in Programmer's Guide
- * [NXPCG-1243] Added comment to README.txt about rolling back to 3.1.
- * [NXPCG-1160] Removed support for online backups and tenant archives, both of which had been previously deprecated.
- * [NXPCG-1192] Increased default value of phraseRecognition.OperationSizeGb to match increase made in SG 3.1 for WB 10.4 ram usage increase.
- * [NXPCG-25] New feature: Disaster Recovery.
- * [NXPCG-332] Media ID Widening - Support long mediaIDs throughout SG3.2
- * [US19912] Removed UpdateMediaPaths tool.
- * [US19912] Removed license dump tool.
- * [US18285] Removed ability to upgrade/migrate from Search Grid

versions prior to 3.1.

- * [US17882] Removed audit reports feature.
- * [FE2130] Removed replica nodes feature.
- * [US14592] Removed support for Linux platform.
- * [US14774] Deprecated out of process decoders. Default setting now disables decoders.
- * [US14775] Deprecated out of process phrase recognition workers. Default setting now disables these workers.
- * [GRID-1290] Added usability enhancements to search console.
- * [US6564] Changed monogest retention setting default to false.
- * [US19334] Removed term collections feature.
- * [DE7738] Fixed control service crash when running in ensemble mode.

```
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==  Changes in 3.2.0 with equivalent changes also in 3.1.22 - 3.1.29  ==
==  -----  ==
==  Provided here for customers who may be upgrading  ==
==  to 3.2 from a version of 3.1 older than 3.1.29.  ==
=====
```

Security Hardening (mostly per Black Duck and Veracode)

- * [NXPCG-1076] Disabled component info (i.e. Jetty version) in HTTP response headers
- * [NXPCG-1067] Updated the following components:
(excludes changes superseded by subsequent changes, above)
 - httpcore to 4.4.15
 - httpmime to 4.5.13
- * [NXPCG-1053] Updated logback to 1.2.10
- * [NXPCG-1035] Updated the following components per Black Duck scan:
(excludes changes superseded by subsequent changes, above)
 - commons-compress to 1.21
 - commons-cli to 1.4 (needed by groovy)
 - commons-codec to 1.15
 - groovy to 2.5.15
 - jetty to 9.4.44.v20210927
 - jline to 2.4.16 (needed by groovy-shell)
 - picocli to 4.3.2 (needed by groovy-console)
 - slf4j to 1.7.32 (to match what logback expects)
- * [NXPCG-1018] Increased size of minimum required Diffie-Hellman key size and disabled potential use of several unsafe TLS algorithms.
- * [NXPCG-1022] Closed JMX ports by default, including its random port RMI opens

in the ephemeral range. Now JMX can be started on-demand if needed without having to restart services Updated Administrator's Guide accordingly.

- * [NXPCG-752] Improved security related to management-console *primitive* mode setting of passwords.
- * [NXPCG-724] Veracode CWE-502 Deserialization of Untrusted Data
- * [NXPCG-665] Veracode CWE-113 Improper Neutralization of CRLF Sequences in HTTP Headers
- * [NXPCG-666] Veracode CWE-73 External Control of File Name or Path
- * [NXPCG-59] Veracode CWE-338 - Use of Cryptographically Weak PRNG
- * [NXPCG-652] Veracode CWE-757 Selection of Less-Secure Algorithm During Negotiation
- * [NXPCG-569] Update java.security to disable TLSv1 and TLSv1.1
- * [NXPCG-733] Updated mxparser to 1.2.2 per Black Duck scan,
- * [NXPCG-360] Updated httpclient to 4.5.13 per Black Duck scan.
- * [US18865] Updated zookeeper to 3.4.14.

----- Bug Fixes

- * [NXPCG-1001] Fixed: Installer UI allowed location and service user changes on upgrades when it shouldn't have.
- * [NXPCG-747] Fixed: Issue with Event Log messages after SG is uninstalled.
- * [NXPCG-690] Fixed: PhrasePackKey equals() method
- * [NXPCG-710] Fixed: DB access race condition with media transactions
- * [NXPCG-108] Fixed: Spurious MIC Failures in SG Java Client
- * [NXPCG-607] Fixed: Corner case service shutdown exception if startup is interrupted
- * [NXPCG-565] Fixed: Idle timeout error messages should be suppressed
- * [NXPCG-358] Fixed: SG 3.2 Programmer's Guide example for XML encoding is incorrect.
- * [NXPCG-339] Fixed: Typos in the Programmer's Guide.
- * [NXPCG-298] Fixed: Property logging on startup is not consistent.
- * [NXPCG-155] Fixed: Workbench error during base service shutdown
- * [NXPCG-82] Fixed: Find Media API should return sooner when nodes are unreachable.

----- General

- * [NXPCG-1260] Add periodic logging of java heap size and overall native memory usage for bwworker processes and all services.
- * [NXPCG-1251] Improved logging of NUMA state
- * [NXPCG-1195] Added displaying platform details to test-system results if new, optional "details" parameter is specified.
- * [NXPCG-1186] Detect SG version mismatches across nodes using "test-system"

- * [NXPCG-1153] Updated and expanded the Administration Guide's, HTTPS support section, corrected and clarified items in the System Requirement's Guide, and made minor tweak to Programmer's Guide related to DR metrics content.
- * [NXPCG-1051] Clarified Admin Guide regarding decommissioning of data nodes.
- * [NXPCG-1047] Updated Admin Guide regarding Control Nodes, particularly regarding deployments with multiple Control Nodes.
- * [NXPCG-753] Improved ResourceManager concurrency (improves structured query performance in most cases)
- * [NXPCG-743] Add Hyper-Threading & NUMA logging at startup
- * [NXPCG-626] Changed our HTML docs to be stand-alone (no assets folder)
- * [NXPCG-624] Validate existence of JRE .jar and .dll files
- * [NXPCG-609] Validate jars in lib against classpath before starting java
- * [NXPCG-605] Enforce jar versioning in wxs files and runtime class paths
- * [NXPCG-301] Add debug logging of node response times for system tests.
- * [NXPCG-298] Improved property logging on start-up.
- * [NXPCG-291] Increased logging clarity for deleteAndGetNext().
- * [NXPCG-290] Added "brief" option for get-systop command.
- * [NXPCG-276] Add logging of invalid search terms to gateway log .
- * [NXPCG-277] Added command-line option to search-console to disable its pre-validation of search terms to allow testing of NXPCG-276.
- * [NXPCG-81] Changed "Beginning media operation" log to info level (was debug)
- * [NXPCG-271] Improve robustness of samples test app against user error.
- * [NXPCG-81] Log media operation beginning info
- * [US19149] Fixed perpetual Base Service crash after digest corruption.
- * [US16882] Updated copyright statement to comply with NiCE copyright notification guidelines.

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